Factuality-Aware Reranking for Extreme Summarization

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Abstract

Abstractive summarizers for XSum often produce fluent but unsupported details. This project studies whether a small reranking pipeline can shift that trade-off toward factuality without redesigning the base generator. I fine-tune `facebook/bart-large-xsum` on a bounded XSum subset, generate beam candidates, and rerank them with generation likelihood, a SummaC-style entailment score, a FactCC-style consistency score, and a lightweight entity-support feature. On a 128-example bounded test split, the final system improves the tracked factuality composite from **0.3324** to **0.4420** relative to the public BART baseline while reducing ROUGE-Lsum from **0.3569** to **0.3398**. A 24-example stratified qualitative analysis set and a bounded MiniCheck check on that same subset both support the direction of the factuality gain. The final result is not a benchmark-scale claim; it is a bounded study showing that simple reranking signals can materially reduce factual errors for extreme summarization at a modest overlap cost.

1 Introduction

Extreme summarization is a good setting for factuality work because the target summary is short, compressed, and easy to drift away from the source article. XSum is especially difficult because its single-sentence references encourage aggressive abstraction rather than extractive copying (Narayan et al., 2018). Strong sequence-to-sequence models such as BART (Lewis et al., 2020) produce fluent outputs on this task, but overlap metrics alone do not guarantee that the generated sentence is supported by the document.

Rather than replacing the generator, I ask whether a bounded reranking layer can make candidate selection more factual. The pipeline fine-tunes BART on XSum, samples multiple beam candidates, scores them with factuality-oriented signals, and selects an operating point that balances ROUGE against factual consistency.

The contribution is not a new summarization architecture. Instead, I leverage a strong public generator and factuality signals from prior work, then test whether a small decision-layer change can improve factual consistency in a measurable and interpretable way. The work therefore emphasizes analysis rather than model count: multi-signal reranking, weight search, one targeted refinement, and qualitative error analysis. The resulting story is conservative: the final system is better on factuality, slightly worse on ROUGE, and still clearly bounded in scale.

2 Background

Prior work motivates both the model choice and the evaluation strategy. BART is a strong baseline for abstractive summarization (Lewis et al., 2020), but factual consistency remains a known failure mode in summarization. FactCC (Kryściński et al., 2020) and SummaC (Laban et al., 2022) motivate two complementary reranking signals: classifier-style factual consistency and entailment-style support from the source document. QAGS (Wang et al., 2020) and FRANK (Pagnoni et al., 2021) further suggest that factuality evaluation benefits from multiple views rather than a single score. More recent work studies both error types and stronger evaluators, including a factual-error taxonomy for summarization (Tang et al., 2023) and MiniCheck for efficient grounded fact-checking (Tang et al., 2024).

Relative to this literature, the goal here is narrower and more practical: use existing factuality signals as reranking features for XSum, then analyze whether that decision-layer change produces a useful factuality-overlap trade-off. The novelty is therefore in the combination of bounded fine-tuning, multi-signal reranking, and analysis-driven refinement rather than in a new metric or generator family.

That framing matters for this project as well. The goal is not to out-scale the literature with a larger model or broader corpus, but to study whether factuality can improve through better selection among candidates produced by a standard summarizer. This keeps the implementation, evaluation, and error analysis tightly connected to one interpretable research question.

3 Method

3.1 Generator and candidate pool

The generator is a bounded fine-tune of `facebook/bart-large-xsum`. Training uses 128 examples from the XSum train split and 64 validation examples for model selection. The best checkpoint (`checkpoint-32`) is exported and used for downstream generation. I then generate candidate pools with beam sizes 4, 8, and 16 on the validation and test splits.

The public `facebook/bart-large-xsum` checkpoint is preserved as the explicit baseline comparator. Baseline selection uses the best-likelihood candidate from the public model; the final system uses the fine-tuned model plus reranking.

3.2 Reranking signals

Each candidate receives four scores:

1. average token log-probability from generation,
2. a SummaC-style support score,
3. a FactCC-style consistency score,
4. an entity-support score derived from named-entity, number, and date precision against the source document.

Search is run over multiple weight settings and beam sizes on the validation split. The selected operating point is `custom_0097` with beam size 16 and weights $(0.0, 0.75, 1.0, 0.5)$ for log-probability, SummaC-style, FactCC-style, and entity support respectively. This choice deliberately prioritizes factuality signals over raw likelihood.

3.3 Evaluation and qualitative analysis

The final test split contains 128 examples. Automatic evaluation reports ROUGE-Lsum (Lin, 2004) plus a factuality composite formed from the tracked SummaC-style, FactCC-style, and entity-support scores. I also use bootstrap confidence intervals over the system deltas.

Metric	Baseline	Final	Delta
ROUGE-Lsum	0.3569	0.3398	-0.0171
Factuality composite	0.3324	0.4420	+0.1096
Qualitative consistent rate	0.0000	0.0417	+0.0417
MiniCheck support rate	0.3333	0.4583	+0.1250

Table 1: Main bounded comparison on the 128-example test split. MiniCheck is reported only on the 24-example qualitative-analysis subset.

For qualitative validation, I review a 24-example stratified sample with explicit provenance (`Codex / AI-assisted expert adjudication`). The sample records the selected system, consistency labels for both summaries, and a primary error type. This is intended for error analysis, not for claims about human inter-annotator reliability.

Finally, I run MiniCheck on the same 24-example subset as bounded supporting evidence, not as a replacement for the main test-set metrics.

4 Results and Discussion

Table 1 shows the main bounded comparison. The final system improves the factuality composite by **+0.1096**, raises MiniCheck support rate on the qualitative-analysis subset by **+0.1250**, and slightly improves the consistent-rate measure on that subset. The cost is a **-0.0171** ROUGE-Lsum drop relative to the public baseline.

The confidence intervals matter. The ROUGE-Lsum delta interval [-0.0310, -0.0029] stays negative, so I cannot claim a quality win on overlap metrics. The factuality delta interval [0.0887, 0.1321] remains positive throughout, so the honest conclusion is a factuality-oriented trade-off rather than a universally better summarizer.

That trade-off is still meaningful in practice. XSum outputs are often judged first on fluency, but a fluent sentence that adds an unsupported entity or relation is still unusable in many downstream settings. The reranker improves the support-oriented metrics without claiming to solve summarization quality in general, so the right interpretation is selective rather than universal.

Figure 1 summarizes the validation-time search frontier. The winning operating point came from a beam-16 search and remains close to the best factuality frontier while keeping ROUGE within a modest distance of the best-likelihood baseline. This is the core result of the project: reranking can push the system toward more supported summaries, but the trade-off is visible and should be reported

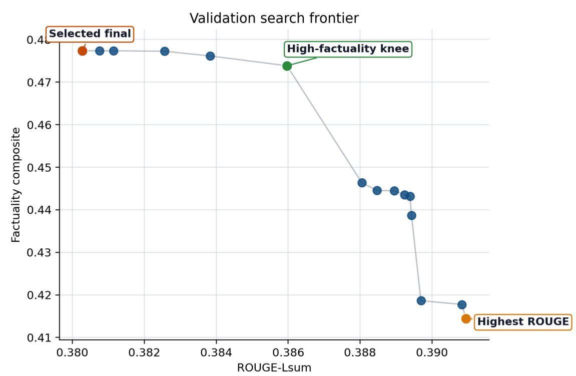

Figure 1: Validation-time search frontier. The selected operating point stays near the best factuality region while preserving a moderate ROUGE-Lsum level.

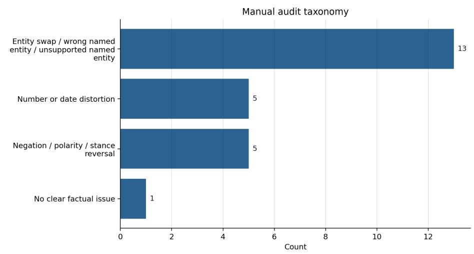

Figure 2: Primary failure types in the 24-example qualitative-analysis subset.

plainly.

4.1 Iterative refinement

After inspecting early failures, I implement entity support as an explicit refinement. The ablation from `logprob_plus_summac_plus_factcc` to `...plus_entity_support` changes factuality composite from 0.4061 to 0.4106 on the bounded test split. This is a small gain, but it is consistent with the error analysis: named-entity substitutions and unsupported entities dominate the remaining failures.

4.2 Qualitative findings

The 24-example qualitative set is intentionally balanced rather than celebratory. The reranker wins 8 examples, the baseline wins 8, and 8 are ties or too close to call. The reranker is still selected more often overall (16/24), and the dominant remaining failure type is entity distortion (12 rows), followed by negation or stance reversal (6) and number/date errors (5).

The qualitative examples show both sides of the trade-off. In article `34001040`, the reranker removes an unsupported motorway detail and keeps the supported claim that further remains were found. In article `40181128`, both systems remain wrong about the named entity, illustrating that reranking alone does not solve entity-level hallucinations once all candidates drift in the same direction.

Figure 2 summarizes the same qualitative subset at the error-type level. Entity distortion is the largest remaining category, which is consistent with both the example-level discussion and the relatively small gain from the entity-support refinement.

4.3 Interpretation

Likelihood alone tends to favor fluent but overcommitted details. Adding entailment-style and FactCC-style signals moves the selected candidates toward more document-supported content, and entity support adds a small but coherent refinement on top.

At the same time, the results are clearly bounded. The run uses 128 training examples, 64 tuning examples, a 128-example test split, and a 24-example qualitative analysis set. Those numbers are large enough to reveal a stable directional trade-off, but not large enough to justify benchmark-scale claims. The qualitative analysis is better suited to identifying failure modes than to making annotator-reliability claims.

The error patterns also clarify where reranking helps and where it stalls. When the candidate set contains both a supported and an unsupported version of the same event, the factuality features often choose the safer sentence. When all beam candidates drift toward the same wrong entity or unsupported relationship, reranking has little room to recover. That helps explain why entity support improves the ablation only modestly and why stronger future improvements likely require better candidate diversity or constrained generation.

5 Conclusion

This bounded XSum study shows that a small factuality-aware reranking layer can materially improve factuality relative to a public BART baseline, even when the generator remains a standard BART summarizer. The final system does not improve every metric: ROUGE-Lsum falls slightly, and entity errors remain the dominant failure mode. The main conclusion is therefore a trade-off claim: candidate reranking is a practical way to trade a modest amount of overlap performance for a meaningful gain in factual consistency.

Limitations

This project is bounded in three important ways. First, the data splits are small relative to full XSum scale, so the results should be interpreted as a bounded study rather than a benchmark claim. Second, the factuality composite uses task-specific support signals rather than a broader panel of external metrics. MiniCheck is included only on the qualitative-analysis subset. Third, the qualitative analysis uses `Codex / AI-assisted expert adjudication` rather than a human-annotator study. These limitations do not remove the main result, but they do constrain how broadly it should be interpreted.

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